LAVA vs. ElasticNet

High-Dimensional Estimation & Links to Causal Forests

Kristian Boroz

May 12, 2026

The High-Dimensional Challenge

- Modern datasets: $p \to n$ or $p \gg n$.
- **OLS** becomes unstable or non-unique.
- Regularisation is essential to control overfitting.
- But the **right** regulariser depends on the **signal regime**:
  1. Sparse: a few large effects.
  2. Dense: many small effects.
  3. **Sparse + Dense**: mixture of both.
- No single classical estimator dominates across all three regimes.

Key Regimes

Regime	Best tool
Sparse	Lasso
Dense	Ridge
Both	**LAVA**

ElasticNet is a compromise between Lasso and Ridge but does **not** *explicitly model a sparse+dense structure.*

Standard ElasticNet: Definition & Objective

ElasticNet merges ℓ_1 (Lasso) and ℓ_2 (Ridge) penalties on the **same** coefficient vector:

$$\widehat{\theta}_{\text{enet}} = \arg\min_{\theta \in \mathbb{R}^p} \left\{ \frac{1}{n}\|Y - X\theta\|_2^2 + \underbrace{\lambda_2\|\theta\|_2^2}_{\text{Ridge term}} + \underbrace{\lambda_1\|\theta\|_1}_{\text{Lasso term}} \right\}$$

Strengths

- Handles correlated features well.
- Produces **sparse** solutions (like Lasso).
- Convex, computationally cheap.
- Single ratio $\alpha = \lambda_1/(\lambda_1+\lambda_2)$ controls the blend.

Limitation

- Both penalties act on the **same** θ: coefficients are either shrunk towards zero *and* kept small so they cannot simultaneously encode a large sparse spike *and* many small non-zero entries.
- Effective ℓ_1 bias grows with λ_1, causing **shrinkage** even in the dense part of the signal.

LAVA: Intuition

LAVA $=$ **L**east **A**bsolute **V**ariations and **A**veraging.

The true parameter is assumed to decompose as

$$\theta_0 = \underbrace{\beta_0}_{\text{Dense component}} + \underbrace{\delta_0}_{\text{Sparse component}}$$

- β_0: many small, non-zero entries $\longrightarrow$ penalised with ℓ_2 (Ridge).
- δ_0: a few large non-zero entries $\longrightarrow$ penalised with ℓ_1 (Lasso).

Key insight

By **splitting** the parameter vector, each component can be regularised *appropriately*, avoiding the over-shrinkage that damages ElasticNet in mixed-signal settings.

LAVA: Objective Function

LAVA jointly estimates the two components (β, δ):

$$(\widehat{\beta}, \widehat{\delta}) = \arg \min_{\beta, \delta \in \mathbb{R}^p} \left\{ \frac{1}{n} \| Y - X(\beta + \delta) \|_2^2 + \lambda_2 \|\beta\|_2^2 + \lambda_1 \|\delta\|_1 \right\}$$

The final predictor is $\widehat{\theta}_{\text{lava}} = \widehat{\beta} + \widehat{\delta}$.

Comparison at a glance

	ElasticNet	**LAVA**
Penalty on	same θ	separate β (Ridge) & δ (Lasso)
Parameters tuned	λ_1, λ_2 (or α, λ)	λ_1, λ_2 independently
Solution type	Sparse	Dense sum of sparse + dense parts
Optimal regime	General correlated features	Sparse + Dense signal

LAVA: Profiled Characterisation

Minimising over β yields a closed-form Ridge solution, so $\widehat{\delta}$ satisfies a **standard Lasso** on *transformed* data:

$$\widehat{\delta} = \arg\min_{\delta \in \mathbb{R}^p} \left\{ \frac{1}{n} \| \widetilde{Y} - \widetilde{X}\,\delta \|_2^2 + \lambda_1 \|\delta\|_1 \right\}$$

where (letting $\mathbf{P}_{\lambda_2} = X(X^\top X + n\lambda_2 I)^{-1} X^\top$ and $\mathbf{K}_{\lambda_2} = I - \mathbf{P}_{\lambda_2}$):

$$\widetilde{Y} = \mathbf{K}_{\lambda_2}^{1/2}\, Y, \qquad \widetilde{X} = \mathbf{K}_{\lambda_2}^{1/2}\, X$$

The full predictor is recovered as:

$$\widehat{\theta}_{\text{lava}} = \mathbf{P}_{\lambda_2} Y + \mathbf{K}_{\lambda_2} X\, \widehat{\delta}$$

LAVA: Bias Reduction & Degrees of Freedom

Post-LAVA debiasing

Once $\widehat{J} = \mathrm{supp}(\widehat{\delta})$ is identified, run OLS on the active sparse set and add the Ridge dense estimate:

$$\widehat{\theta}_{\text{post-lava}} = \widehat{\beta} + \widetilde{\delta}_{\text{OLS on } \widehat{J}}$$

This **removes** the ℓ_1 shrinkage bias from the sparse component while retaining the dense signal.

Degrees of freedom (SURE)

$$df = \mathbb{E}\left[\mathrm{rank}(\widetilde{X}_{\widehat{J}}) + \mathrm{tr}(\widetilde{\mathbf{K}}_{\widehat{J}}\mathbf{P}_{\lambda_2})\right]$$

Unlike Lasso (where $df = |\widehat{J}|$), LAVA's complexity **accounts for both** the selected sparse features *and* the dense Ridge projection, giving more honest model-selection criteria.

LAVA: Takeaway lesson

Practical implication

LAVA can be implemented as a two-step procedure:

(1) Ridge-project the data

(2) run Lasso on residuals

Tuning (λ_1, λ_2) is done via `GridSearchCV` or SURE.

Performance guarantee

LAVA achieves **vanishing prediction error**

whenever the transformed-data Lasso is consistent

even when neither a pure sparse nor a pure dense estimator would succeed alone.

Side-by-Side Comparison

Feature	ElasticNet	LAVA
Penalty structure	$\ell_1 + \ell_2$ on same θ	ℓ_2 on β, ℓ_1 on δ separately
Solution type	Sparse	Dense ($\widehat{\beta} + \widehat{\delta}$)
Bias in dense regime	High (ℓ_1 shrinks small coefficients)	Low (Ridge handles dense part)
Bias in sparse regime	Medium	Low (post-LAVA debiasing)
Optimal regime	Correlated features / generally sparse	Sparse + Dense mixture
Tuning	α (ratio) + λ	λ_1, λ_2 independently
Computational cost	Low (coordinate descent)	Moderate (grid search or SURE)
Degrees of freedom	$\lvert\widehat{J}\rvert$ (non-zeros)	Accounts for both Ridge & Lasso

The Double Machine Learning Pipeline

Causal Forests estimate Conditional Average Treatment Effects (CATE): $\tau(x) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = x]$.

They rely on **Robinson's Partialling-Out Transformation**:

$$\underbrace{(Y_i - m(X_i))}_{\widetilde{Y}_i} = \tau(X_i)\,\underbrace{(W_i - e(X_i))}_{\widetilde{W}_i} + \varepsilon_i$$

Nuisance functions (Stage 1)

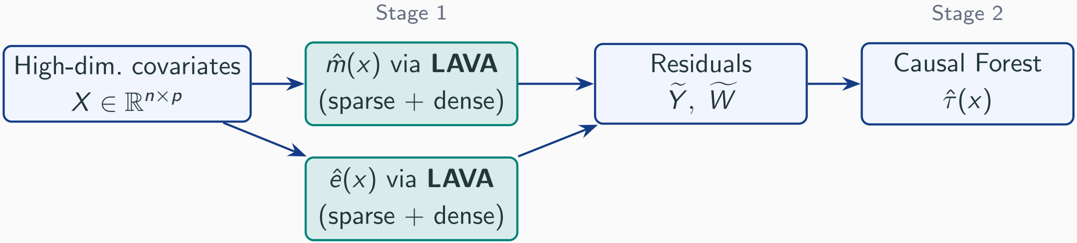
- $m(x) = \mathbb{E}[Y \mid X = x]$: outcome model.
- $e(x) = \mathbb{E}[W \mid X = x]$: propensity score.
- Both must be estimated **accurately** to avoid bias in $\hat{\tau}(x)$.

CATE estimation (Stage 2)

Causal Forest regresses $\widetilde{Y}_i$ on $\widetilde{W}_i$ using **adaptive forest weights** $\alpha_i(x)$:

$$\hat{\tau}(x) \sim \sum_i \alpha_i(x)\, \widetilde{Y}_i \widetilde{W}_i / \widetilde{W}_i^2$$

Where LAVA Fits In



- **Why LAVA here?** In high-dimensional observational data, $m(x)$ and $e(x)$ often involve a mix of a few dominant confounders (sparse) and many weak background effects (dense) — precisely LAVA's sweet spot.
- Better residuals $\widetilde{Y}, \widetilde{W} \Rightarrow$ **Double Robustness** is satisfied $\Rightarrow$ **Regularisation Bias** does *not* bleed into $\hat{\tau}(x)$.

Regularisation Bias: The Hidden Danger

If Stage 1 uses a *misspecified* or *over-shrunken* nuisance model:

$$\underbrace{\hat{m}(x)}_{\text{biased by } \ell_1} \quad \Rightarrow \quad \widetilde{Y}_i = Y_i - \hat{m}(X_i) \text{ contains confounding residual}$$

$$\Rightarrow \quad \hat{\tau}(x) \text{ is spuriously heterogeneous}$$

ElasticNet (dense regime)

Over-penalises many small confounders $\Rightarrow$ residual confounding leaks into $\widetilde{Y}$ $\Rightarrow$ **false treatment heterogeneity** can be "hallucinated."

LAVA (sparse + dense regime)

Correctly absorbs both large confounders (Lasso) and many small ones (Ridge) $\Rightarrow$ cleaner residuals $\Rightarrow$ **honest CATE estimates**.

This is why Chernozhukov et al. (2018) and Wager & Athey (2018) both stress cross-fitting and flexible nuisance estimators: the quality of $\hat{\tau}$ is only as good as the quality of Stage 1.

Key Takeaways

1. **ElasticNet** blends Lasso and Ridge penalties on a *single* coefficient vector — effective for correlated features but cannot cleanly separate sparse and dense signal components.
2. **LAVA splits** the parameter into a dense part (ℓ_2) and a sparse part (ℓ_1), dominates in the "Sparse + Dense" regime and achieves lower prediction error than either Lasso or Ridge alone.
3. The **profiled** form of LAVA reduces to a Lasso on Ridge-transformed data — making it implementable with standard tools.
4. In **Causal Forests** using LAVA for nuisance estimation of $m(x)$ and $e(x)$ ensures double robustness and prevents regularisation bias from distorting CATE estimates.
5. **Rule of thumb:** If you suspect both a handful of strong predictors *and* a background of many weak ones, prefer LAVA over ElasticNet — especially when downstream causal inference is the goal.